

Kneeling tolerance when using quadriceps tendon autograft for anterior cruciate ligament reconstruction is superior to hamstring tendon autograft

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Abstract

Purpose: To investigate kneeling tolerance in patients undergoing hamstring (HT) versus quadriceps (QT) anterior cruciate ligament reconstruction (ACLR) and investigate correlation with patient-reported outcome measures (PROMs).

Methods: After recruitment and randomisation, 112 patients (HT = 55; QT = 57) underwent ACLR. Patients were assessed at 6, 12 and 24 months using the Kneeling Tolerance Test, which evaluates patient-reported pain in a position of both 90 (KT90) and 110 (KT110) degrees of knee flexion. PROMs collected included the International Knee Documentation Committee (IKDC) questionnaire and the ACL Return to Sport after Injury (ACL-RSI) questionnaire.

Results: Kneeling tolerance at KT90 and KT110 improved ($p < 0.05$) for both graft types across all time points. There was no difference in KT90 scores between groups at 6 or 12 months. At 24 months, kneeling tolerance was superior in the QT group (mean HT 93 ± 9 vs. QT 98 ± 5 ; $p = 0.003$). For KT110 scores, a statistically significant difference was noted at 6 (mean HT 80 ± 25 vs. QT 89 ± 12 ; $p = 0.027$), 12 (mean HT 90 ± 13 vs. QT 95 ± 10 ; $p = 0.040$) and 24 months (mean HT 92 ± 10 vs. QT 97 ± 5 ; $p = 0.003$). The ACL-RSI was significantly correlated with KT90 and KT110 at 24 months ($r = 0.40$, $p < 0.001$; $r = 0.40$, $p < 0.001$). Other PROMs demonstrated significant weak-to-moderate correlations with kneeling tolerance.

Conclusion: Patients undergoing ACLR with a QT versus HT autograft report superior kneeling tolerance up to 2 years postsurgery, more prominent in deeper (110°) knee flexion. A strong correlation with ACL-RSI was demonstrated at 2 years.

Registration: ACTRN12618001520224p (Australian New Zealand Clinical Trials Registry).

Level of Evidence: Level 1.

KEYWORDS

anterior cruciate ligament reconstruction, hamstrings autograft, kneeling, quadriceps autograft

Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; ACL-RSI, Anterior Cruciate Ligament Return to Sport after Injury; BPTB, bone–patella tendon–bone; GRC, Global Rating of Change; HT, hamstring tendon; IKDC, International Knee Documentation Committee; KT110, kneeling tolerance at 110 degrees; KT90, kneeling tolerance at 90 degrees; LKS, Lysholm Knee Score; PROM, patient-reported outcome measure; QT, quadriceps tendon; RCT, randomised control trial; ROM, range of motion.

INTRODUCTION

Anterior cruciate ligament (ACL) tears are common, with surgical reconstruction (ACLR) considered the current standard clinical treatment [24] with non-operative management considered in select patients. In Australia, the incidence of ACLR is the highest in the world with a previously reported annual incidence of 77.4 per 100,000 [31]. This injury predominantly affects young adults [31], a sub-population in which kneeling ability plays an important role in knee function for sporting and occupational activities. A variety of grafts have been described; however, the most common are hamstring tendons (HT) and bone–patella tendon–bone (BPTB) grafts [19], with increasing interest being shown for quadriceps tendon (QT) [27].

Regarding the functional outcome of HT versus QT grafts, there are five level III studies [6, 8, 17, 20, 23, 26], four level II studies [10, 11, 14, 16] and four level I studies [1, 2, 12, 25]. However, none of these trials have directly compared the kneeling tolerance of these two patient groups. The authors have previously shown that 77% of HT graft patients report kneeling difficulty at 1 year and that this correlates with clinical outcome measures [4]. Pinczewski et al. [21] found that 27% of their HT graft patients had pain on kneeling at 10 years. This is in comparison to Han et al. [9] who found only 5% of QT graft patients had pain on kneeling. Given the relative lack of information available on kneeling capacity after ACLR, and the need for many to kneel daily for various daily activities, this study aimed to investigate whether there was a quantifiable difference between the kneeling tolerance observed in patients undergoing either HT or QT autograft. It was hypothesised that (1) there would be no statistically significant difference between HT and QT cohorts in kneeling tolerance and (2) strong and significant correlations would be observed between kneeling tolerance and commonly reported patient-reported outcome measures (PROMs).

MATERIALS AND METHODS

Participants and recruitment

Patients undergoing ACLR via a single surgeon (R.R.), with or without concomitant meniscal pathology, were recruited for the study. The study was conducted as per the methods described in a previous publication of the preliminary results of this trial [7]. This parallel-group study randomized patients to either an HT ($n=57$) or QT ($n=59$) autograft construct using a 'random number sequence generator'. The random number sequence was maintained by an independent researcher, with each group randomization released upon recruitment and consent. Patients were included if they were aged 16 to 50 years and qualified for ACLR

based on clinical examination and magnetic resonance imaging. The exclusion criteria were as follows: unable or unwilling to sign the patient informed consent form; a body mass index of ≥ 40 ; revision or multiligamentous reconstruction; presenting with significant articular cartilage pathology (Kellgren–Lawrence grade ≥ 3) and being treated for a psychiatric disorder or the presence of alcohol or substance abuse. Ethical approval was provided by the relevant hospital Human Research Ethics Committee, and the trial was conducted according to the Declaration of Helsinki and CONSORT guidelines. The trial was registered in the Australian New Zealand Clinical Trials Registry.

ACLR surgical techniques

All ACLR procedures were conducted by a single surgeon with patients under general anaesthesia. A standardized anaesthetic protocol was followed for all patients, including a single-shot adductor canal block using 0.375% ropivacaine and 8 mg dexamethasone, administered under ultrasound guidance after the patient was anaesthetized.

In the HT group, the graft was harvested through a transverse incision over the pes anserinus using a closed tendon harvester to obtain the semitendinosus, which was quadrupled and prepared. The Arthrex Graftlink system (Arthrex) was used, allowing for a single HT harvest in over 90% of cases. In the remaining cases, the gracilis tendon was also harvested to ensure a minimum graft diameter of 8 mm in all patients. For the QT group, an all-soft tissue graft was taken via a longitudinal incision over the distal quadriceps tendon and the proximal patella. A 9-, 10- or 11-mm graft harvester was used, depending on patient size, to harvest approximately 70 mm of graft tissue, which was prepared according to the Arthrex Graftlink system guidelines.

Patients underwent knee ligament examination under anaesthesia, followed by diagnostic arthroscopy to assess meniscal or chondral damage, with surgery performed if necessary. Any additional procedures (e.g., meniscectomy, meniscal repair, simple chondroplasty) were carried out before ACLR. Femoral and tibial tunnels were prepared for graft passage in an antegrade fashion, using adjustable loop fixation for both the femoral and tibial sides. Femoral tunnel preparation was done using an inside–out technique, drilling through the anteromedial portal for a line-to-line fit. Graft tensioning was performed with the knee in full extension, according to the manufacturer's recommendations, and the graft was cycled before final fixation. Wounds were closed in layers, haemostasis was confirmed and absorbable subcuticular sutures were used. Dressings were applied, followed by wool and crepe bandages. Postoperative analgesia included regular paracetamol, non-steroidal anti-inflammatories

and stronger pain relief for breakthrough pain. Patients also received standardized education on icing, compression and limb elevation to manage swelling.

Rehabilitation

Early rehabilitation during the first 2 weeks focused on reducing pain and swelling, ensuring safe and proper use of crutches, and performing exercises to improve patellar mobility, restore early knee range of motion (ROM) and regain quadriceps control. For patients who also had meniscal repair, their weight-bearing status was adjusted, with a gradual reduction in crutch use and bracing over the first 6 weeks. Specifically, patients with unstable meniscal tears (such as radial tears) followed a restricted weight-bearing protocol for up to 6 weeks, while those with stable tears were allowed to bear weight immediately after the repair, as tolerated. Early hydrotherapy was recommended for all patients after the 2-week orthopaedic review, provided the surgical incisions had healed. Although the first 6 weeks of rehabilitation were relatively conservative to manage pain and swelling, progressive strengthening exercises were introduced 6 weeks postsurgery. Rehabilitation during the initial 6 weeks was not standardized and was guided by the patient's outpatient physical therapist.

Clinical assessment

Kneeling tolerance was assessed using the previously validated Kneeling Tolerance Test [5]. This test assesses kneeling tolerance at 90 (KT90, to assess loading through the patellofemoral joint) and 110 (KT110, to assess tibial loading) degrees of flexion. Both positions are assessed on three surfaces: hard, firm and soft. Therefore, a score is obtained for each of the two knee positions (KT90 and KT110), expressed as a percentage of kneeling tolerance (i.e., a score of 100 represents pain-free kneeling on all three surfaces). Participants underwent testing at 6, 12 and 24 months. All clinical reviews were conducted within the same private outpatient physical therapy clinic.

Patients also completed a range of PROMs, including (1) the International Knee Documentation Committee (IKDC) Subjective Knee Form [13], (2) the Lysholm Knee Rating Scale (LKS) [18] and (3) the ACL Return to Sport after Injury (ACL-RSI) score [29]. Finally, a Global Rating of Change (GRC) scale [15] evaluated the patients' perceived status compared to before their surgery.

Data and statistical analysis

The data for this study is a secondary analysis of a prospective randomized controlled trial (RCT). A priori

power calculation was determined based on the primary outcome variable of the primary study (anterior translation via the KT-1000 arthrometer) between the two surgical groups. The study was powered to detect a minimum 1 mm difference in laxity, adapted from a previously published study [10]. Therefore, 102 patients (51 in each group) were required to reveal differences at the 5% significance level, with a power of 80%.

Independent sample *t*-tests were employed to compare KT90 and KT110 Knee Tolerance Test means between the HT and QT groups at each of the 6-, 12- and 24-month time points after checking for normality. Spearman's correlation coefficients were used to assess the association between kneeling tolerance and PROMs. Interpretation of the size of observed correlation coefficients was guided by standard convention as per Dancey and Reidy [3]. Associations were assessed as very strong (>0.70), strong (0.40–0.69), moderate (0.30–0.39), weak (0.20–0.29) or negligible (0.01–0.19). Statistical analysis was performed using SPSS software Version 27.0 (SPSS Inc.), with significance determined at $p < 0.05$.

RESULTS

Between March 2019 and December 2020, a total of 116 patients were recruited and randomized for the study (Table 1). One hundred and twelve patients proceeded to surgery (Figure 1). Of the 112 patients (HT = 55; QT = 57) who were recruited and underwent surgery, 97 patients (HT = 49; QT = 48) were assessed at the 24-month review. Patient-reported outcome measures are listed in Table 2.

Kneeling tolerance

At 6 months postsurgery, all participants had achieved sufficient knee flexion to allow assessment of knee

TABLE 1 Characteristics of patients in the hamstring (HT) and quadriceps (QT) tendon autograft groups who were recruited, randomized and underwent surgery.

Variable	Measure	HT (<i>n</i> = 55)	QT (<i>n</i> = 57)
Age, years	Mean $\pm$ SD	29.4 $\pm$ 7.7	28.1 $\pm$ 8.2
Body mass index	Mean $\pm$ SD	26.3 $\pm$ 3.6	26.6 $\pm$ 3.6
Injury to surgery time, weeks	Mean $\pm$ SD	9.6 $\pm$ 8.7	8.5 $\pm$ 8.8
Sex, male	<i>n</i> (%)	28 (50.9)	28 (49.1)
Injury mechanism, noncontact	<i>n</i> (%)	44 (80.0)	46 (80.7)
Concomitant surgery	<i>n</i> (%)	34 (61.8)	32 (52.6)
Meniscectomy	<i>n</i> (%)	2 (3.6)	0 (0.0)
Meniscal repair	<i>n</i> (%)	32 (58.2)	32 (52.6)

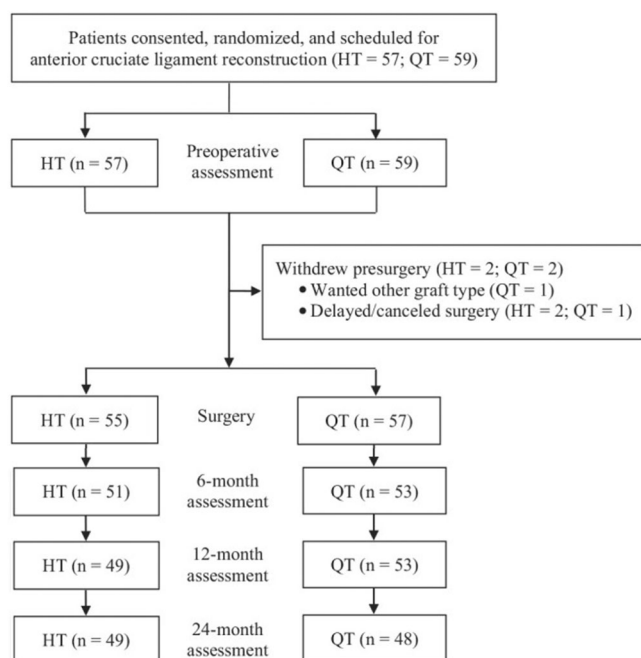


FIGURE 1 Flowchart demonstrating patient recruitment and evaluation over the 24-month postoperative period in patients randomized to the HT and QT groups. HT, hamstring; QT, quadriceps.

TABLE 2 Patient-reported outcome measures in the hamstring (HT) and quadriceps (QT) tendon autograft groups.

	6 months		12 Months		24 months	
	HT	QT	HT	QT	HT	QT
IKDC	74 ± 12	72 ± 15	86 ± 11	83 ± 14	92 ± 8	89 ± 10
LKS	85 ± 11	82 ± 12	90 ± 9	89 ± 11	94 ± 7	93 ± 8
ACL-RSI	54 ± 24	48 ± 26	69 ± 25	61 ± 25	73 ± 27	73 ± 22
GRC	2 ± 2	2 ± 2	4 ± 1	3 ± 2	4 ± 1	4 ± 1

Note: Values are presented as mean ± standard deviation.

Abbreviations: ACL-RSI, ACL Return to Sport after Injury score; GRS, Global Rating of Change scale; IKDC, International Knee Documentation Committee Subjective Knee Form; LKS, Lysholm Knee Rating Scale.

flexion at the two flexion positions. There was no difference evident in KT90 scores (Figure 2) between groups at 6 or 12 months. At 24 months postsurgery, kneeling tolerance was superior in the QT group (mean HT 93 ± 9 vs. QT 98 ± 5; $p = 0.003$). For KT110 scores (Figure 3), a statistically significant difference was noted at 6 (mean HT 80 ± 25 vs. QT 89 ± 12; $p = 0.027$), 12 (mean HT 90 ± 13 vs. QT 95 ± 10; $p = 0.040$) and 24 months (mean HT 92 ± 10 vs. QT 97 ± 5; $p = 0.003$).

Correlations: Kneeling tolerance and PROMs

Variable correlations were demonstrated across time points (Table 3). IKDC scores demonstrated a weak

correlation with both KT90 and KT110 at 6 months, increasing to moderate at 12 and 24 months (Spearman KT90 = 0.26, 0.36, 0.31; KT110 = 0.26, 0.36, 0.30). Lysholm scores demonstrated moderate correlations with KT90 and KT110 at all time points, except a weak correlation with KT110 by 24 months (Spearman KT90 = 0.33, 0.31, 0.30; KT110 = 0.33, 0.35, 0.29). ACL-RSI demonstrated a negligible correlation at 6 months with an increase to strong by 24 months (Spearman KT90 = 0.13, 0.27, 0.40; KT110 = 0.13, 0.33, 0.40). Correlation with GRC remained weak through all time points (Spearman KT90 = 0.20, 0.26, 0.24; KT110 = 0.20, 0.26, 0.24).

DISCUSSION

The most important finding of the present study was that patient-reported kneeling tolerance, at least in a more flexed (110°) knee position, was superior in patients undergoing ACLR with a QT (vs. HT) autograft. The ability to kneel is important in undertaking several activities of daily living, as well as numerous religious, social and occupational tasks. Pain, symptoms, patient satisfaction and the return of other functional abilities are commonly evaluated after various lower limb orthopaedic surgeries and are often reported as key components of the success of the surgical procedure. However, despite the importance of kneeling ability and the anecdotal dissatisfaction with this capability often reported by patients following ACLR, there is still a lack of data on this topic. This is the first paper to directly investigate kneeling tolerance objectively in QT and HT autografts for ACLR.

This study demonstrated that kneeling tolerance improves with time postoperatively, with continued improvement demonstrated late in the recovery. In the validation of the Kneeling Tolerance Test [5], the authors have previously demonstrated a mean KT90 score of 95 and KT110 score of 93 in an asymptomatic population. Taking this into account, it would be reasonable to conclude that those undergoing a QT graft will be more likely to reach 'normal' kneeling tolerance at 12 months and those undergoing HT graft will continue to have a slight reduction in kneeling tolerance at 24 months. Whether this slight reduction is clinically significant is uncertain. Nonetheless, the statistically significant difference between groups in the current study did not support the first hypothesis.

The difference demonstrated between 90° and 110° is likely due to the shift of load bearing when kneeling in these positions. It has been previously shown that at 90°, load bearing is through the surface overlying the patella, with this shifting to around the tibial tubercle at 110° [30]. The use of a QT graft may provide improved kneeling tolerance at 110° by avoiding damage to the large sensory nerves of the infrapatellar branch of the

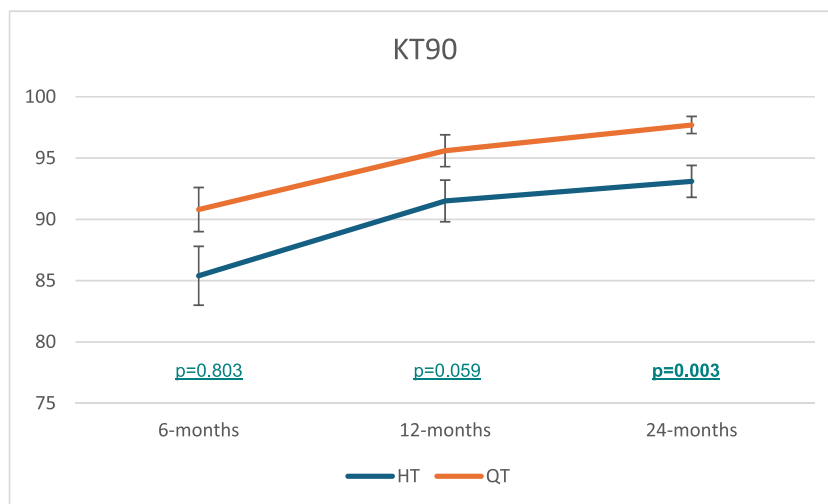


FIGURE 2 Mean Kneeling Tolerance Test scores for patients in the hamstring (HT) and quadriceps (QT) tendon autograft groups over the 24-month period, in the 90° knee flexion position (KT90). Error bars represent standard error.

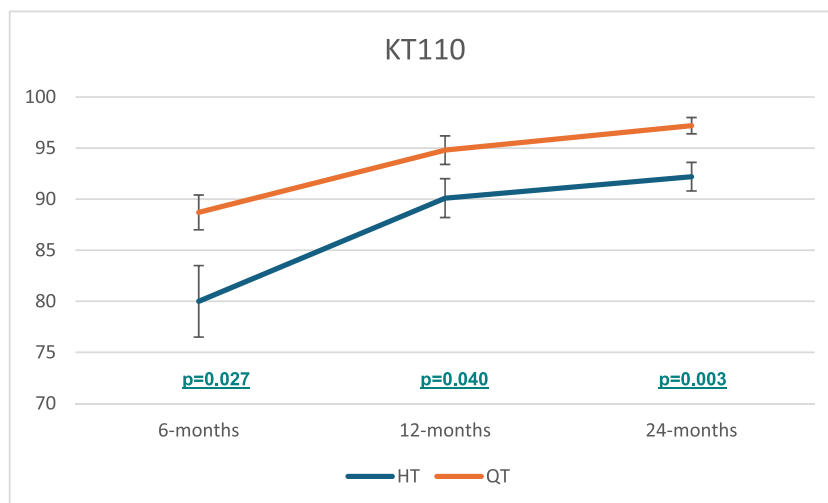


FIGURE 3 Mean Kneeling Tolerance Test scores for patients in the hamstring (HT) and quadriceps (QT) tendon autograft groups over the 24-month period, in the 100° knee flexion position (KT110). Error bars represent standard error.

TABLE 3 Pearson correlations between the Kneeling Tolerance Test and patient-reported outcome measures.

	6 Months		12 Months		24 Months	
	KT90	KT110	KT90	KT110	KT90	KT110
IKDC	0.26 ^a	0.26 ^a	0.36 ^a	0.36 ^a	0.31 ^a	0.3 ^a
LKS	0.33 ^a	0.33 ^a	0.31 ^a	0.35 ^a	0.3 ^a	0.29 ^a
ACL-RSI	0.13	0.13	0.27 ^a	0.33 ^a	0.4 ^a	0.4 ^a
GRC	0.2 ^b	0.2 ^b	0.26 ^a	0.26 ^a	0.24 ^b	0.24 ^b

Abbreviations: ACL-RSI, ACL Return to Sport after Injury score; GRS, Global Rating of Change scale; IKDC, International Knee Documentation Committee Subjective Knee Form; KT90, Kneeling Tolerance 90; KT110, Kneeling Tolerance 110; LKS, Lysholm Knee Rating Scale.

^aCorrelation is significant at the 0.01 level (two-tailed).

^bCorrelation is significant at the 0.05 level (two-tailed).

saphenous nerve (IPBSN). In comparison, HT autograft harvest commonly results in pre-tibial paresthesia, with Portland et al. [22] finding a self-reported prevalence of 43%–59%, depending on the orientation of the incision used for hamstring harvest.

The reported prevalence of kneeling difficulties following ACLR varies significantly in the literature. The largest study [21] assessing kneeling pain involved 180 patients who underwent either patellar or hamstring autografts. In this study, only 6% of patients reported kneeling pain at 2 years postsurgery, but this number increased dramatically to 29% at 5 years. The researchers used a simple visual analogue scale (0–10) to measure pain severity but did not explore the reasons behind the pain or whether symptoms were also

present in the unaffected knee. In contrast, a smaller study by Webster et al. [28] found a higher incidence of 62% at 4 months, which dropped to 25% at 2 years, but then increased to 41% at 15 years. The authors have previously reported on the prevalence of kneeling difficulty in those undergoing reconstruction with an HT graft [4]. Only 23% of patients reported no kneeling difficulty or discomfort at 1 year, with this figure improving to 46% at 2 years. Among those having trouble, most reported limitations in the duration they could kneel at 1 year. By 2 years, however, the majority stated that while they could kneel with some pain, there were no longer any limitations on duration.

The IKDC includes a single question related to kneeling ability, while the LKS and ACL-RSI do not. The second hypothesis was partially supported, and the association found between kneeling tolerance and PROMs (even those that do not specifically address kneeling tolerance) suggests that kneeling difficulty may be an underappreciated concern for patients, warranting further investigation. It may also be the case that existing PROMs do not adequately capture the importance of kneeling difficulty on the overall function of the knee and may warrant the development of more targeted outcome measures. The moderate to strong associations observed at certain time points for most scores, except the GRC, indicate that kneeling might impact a patient's self-perceived function, quality of life and postsurgery improvement. Exploring the underlying causes of kneeling difficulties could lead to adjustments in surgical techniques that enhance this aspect of kneeling function. Additionally, for patients experiencing kneeling difficulties, targeted rehabilitation programs aimed at improving kneeling tolerance may enhance their perceived function and overall quality of life.

Several limitations of this study should be acknowledged. First, although the RCT achieved a strong retention rate and included comprehensive subjective and objective assessments over the 24-month period, it was not double-blinded. Patients were aware of their group assignments, and the differing scar locations between the two groups made it difficult to blind the physical therapists conducting the assessments. Second, the quality of rehabilitation can influence strength recovery and functional outcomes. While a general rehabilitation plan was provided, this was a community-based cohort, with patients receiving rehabilitation guidance and return-to-sport (RTS) advice from various physical therapists. Additionally, there was no standardized preoperative rehabilitation period, which is now recommended as part of the standard protocol for patients undergoing ACL reconstruction. Lastly, the absence of prior data made power calculation difficult. Instead, the power calculation from the primary study was used.

CONCLUSION

The current study demonstrated that patient-reported kneeling tolerance, at least in a more flexed (110-degrees) knee position, was superior in patients undergoing ACLR with a QT (vs. HT) autograft for up to 2 years. However, both groups showed improvement with time, and by 2 years, kneeling tolerance was high in both groups with minimal deficit. Apart from a strong correlation between kneeling tolerance and the ACL-RSI at 2 years, the correlation between kneeling tolerance and other PROMs, across all postoperative time points, was weak to moderate. This information may assist clinicians in tailoring the choice of graft to their patient based on their functional daily, occupational and sporting demands.

AUTHOR CONTRIBUTIONS

The following authors have conceived and designed the study (Jay R. Ebert, Nicholas D. Calvert, Ross Radic), supervised the conduct of the study (Jay R. Ebert, Nicholas D. Calvert, Ross Radic), analyzed the data (Jay R. Ebert and Nicholas D. Calvert), wrote the initial drafts (Jay R. Ebert and Nicholas D. Calvert), critically revised the manuscript (Jay R. Ebert, Nicholas D. Calvert, and Ross Radic) and ensure the accuracy of the data and analysis (Jay R. Ebert, Nicholas D. Calvert, and Ross Radic). The author confirms that all authors have seen and agreed with the contents of the manuscript and agrees that this work has not been submitted or published elsewhere in whole or part.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data has not been made publicly available, though data sets generated during the current study can be made available from the corresponding author on reasonable request.

ETHICS STATEMENT

Ethics approval was obtained by the Hollywood Private Hospital (HPH541). The trial was registered in the Australian New Zealand Clinical Trials Registry (ACTRN12618001520224p). Informed and written consent was obtained from all individual participants included in the study.

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